

Final project: Research Plan

1. Understand the Datasets

- **Orders** → Tracks customer purchases (order ID, user ID, restaurant ID, timestamps, price, quantity, etc.).
- **Users** → Provides customer details (location, demographics, etc.) to help segment demand.
- **Restaurants** → Contains details about each restaurant (name, cuisine type, location, rating, etc.).
- **Food/Menu** → Lists food items with prices and the restaurant menus they belong to.

2. Data Cleaning & Preparation

- Standardize keys (restaurant_id, food_id, user_id, etc.) across tables for joining.
- Handle missing values (e.g., missing user demographics or incomplete menu prices).
- Remove duplicates (common in orders datasets).
- Convert prices, dates, and ratings into consistent formats.

3. Data Integration

- Join **orders** with **restaurants** → to know which restaurant each order came from.
- Join **orders** with **menu/food** → to analyze sales by item, category, or cuisine.
- Join **orders** with **users** → to find customer preferences by location, age group, etc.

4. Key Analysis Questions

Restaurant Performance

- Which restaurants generate the **highest revenue**?
- Which restaurants have the **highest number of orders**?
- What's the **average order value (AOV)** per restaurant?
- Which restaurants are **growing** (order volume/revenue trend over time)?
- **Cuisine Contribution to Total Revenue** – Which cuisines account for the largest revenue share?
- **Highest Rated Cuisines** – Which cuisines consistently get top ratings?
- **Lowest Rated Cuisines** – Where do customers tend to be less satisfied?
- Do **higher-rated restaurants** receive **more orders** on average?

5. Visualization & Insights (Answering the Questions)

- **Highest Revenue Restaurants** → Bar chart of top 10 restaurants by revenue.
- **Most Ordered Restaurants** → Column chart showing restaurants with the highest order counts.
- **Average Order Value (AOV)** → Box plot or bar chart comparing AOV across restaurants.
- **Growth Over Time** → Line chart of order volume/revenue trend per restaurant.

- **Cuisine Revenue Share** → Pie chart or stacked bar showing cuisine contribution to total revenue.
- **Cuisine Ratings** → Heatmap comparing cuisines by average rating vs. number of orders.
- **High vs. Low Ratings Impact** → Scatterplot showing ratings on the x-axis vs. order volume/revenue on the y-axis.

6. Interpretation (Explaining the Results)

- **Revenue Leaders** → Identify why certain restaurants dominate revenue (e.g., location, cuisine type, large menu).
- **Customer Demand Drivers** → Show how ratings, menu variety, and cuisine popularity affect customer choices.
- **Growth Patterns** → Explain which restaurants/cuisines are trending and why (seasonality, promotions, delivery area expansion).
- **Loyalty Effect** → Highlight cuisines/restaurants that retain customers (high repeat orders).
- **Rating Impact** → Interpret whether high ratings consistently lead to higher sales, or if some cuisines succeed even with lower ratings (e.g., fast food).

7. Deliverables (Final Output)

- **Cleaned Dataset** → Standardized, deduplicated, and ready for analysis.
- **Interactive Dashboard (Tableau/Power BI/Excel)** with KPIs:
 - Revenue by restaurant
 - Orders by restaurant & unique customers
 - Cuisine contribution to revenue & ratings
 - Growth trend of top-performing restaurants
- **Written Insights & Recommendations** → Summary highlighting:
 - Which restaurants/cuisines are top performers
 - Why they succeed (ratings, loyalty, menu size, location)
 - Strategic opportunities (e.g., cuisines with high ratings but low supply, underperforming areas with high demand potential).